

DEFINE & TEST

JSS 1-3 | Soilless Learning Garden

Use scientific investigation to define a question about recirculating soilless operation. Quantify the response and defend a recommendation.

MISSION KIT system map, pH/EC/flow tools

EVIDENCE solution, flow or root-zone evidence

GM-371 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Define with system map, pH/EC/flow tools. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for solution, flow or root-zone evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Teacher supervises solution measurements. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Soilless Learning Garden, including the evidence that triggers the decision.

SOILLESS SCIENCE MISSION

JSS 1-3 | Soilless Learning Garden

Use scientific investigation to hypothesize a question about recirculating soilless operation. Quantify the response and defend a recommendation.

MISSION KIT system map, pH/EC/flow tools

EVIDENCE solution, flow or root-zone evidence

GM-372 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Hypothesize with system map, pH/EC/flow tools. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for solution, flow or root-zone evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Teacher supervises solution measurements. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Soilless Learning Garden, including the evidence that triggers the decision.

DESIGN THE EVIDENCE

JSS 1-3 | Soilless Learning Garden

Use scientific investigation to design a question about recirculating soilless operation. Quantify the response and defend a recommendation.

MISSION KIT system map, pH/EC/flow tools

EVIDENCE solution, flow or root-zone evidence

GM-373 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Design with system map, pH/EC/flow tools. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for solution, flow or root-zone evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Teacher supervises solution measurements. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Soilless Learning Garden, including the evidence that triggers the decision.

REPLICATE & TEST

JSS 1-3 | Soilless Learning Garden

Use scientific investigation to replicate a question about recirculating soilless operation. Quantify the response and defend a recommendation.

MISSION KIT system map, pH/EC/flow tools

EVIDENCE solution, flow or root-zone evidence

GM-374 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Replicate with system map, pH/EC/flow tools. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for solution, flow or root-zone evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Teacher supervises solution measurements. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Soilless Learning Garden, including the evidence that triggers the decision.

SOILLESS SCIENCE MISSION

JSS 1-3 | Soilless Learning Garden

Use scientific investigation to quantify a question about recirculating soilless operation. Quantify the response and defend a recommendation.

MISSION KIT system map, pH/EC/flow tools

EVIDENCE solution, flow or root-zone evidence

GM-375 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Quantify with system map, pH/EC/flow tools. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for solution, flow or root-zone evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Teacher supervises solution measurements. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Soilless Learning Garden, including the evidence that triggers the decision.

ANALYZE THE EVIDENCE

JSS 1-3 | Soilless Learning Garden

Use scientific investigation to analyze a question about recirculating soilless operation. Quantify the response and defend a recommendation.

MISSION KIT system map, pH/EC/flow tools

EVIDENCE solution, flow or root-zone evidence

GM-376 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Analyze with system map, pH/EC/flow tools. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for solution, flow or root-zone evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Teacher supervises solution measurements. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Soilless Learning Garden, including the evidence that triggers the decision.

DIAGNOSE & TEST

JSS 1-3 | Soilless Learning Garden

Use scientific investigation to diagnose a question about recirculating soilless operation. Quantify the response and defend a recommendation.

MISSION KIT system map, pH/EC/flow tools

EVIDENCE solution, flow or root-zone evidence

GM-377 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Diagnose with system map, pH/EC/flow tools. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for solution, flow or root-zone evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Teacher supervises solution measurements. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Soilless Learning Garden, including the evidence that triggers the decision.

SOILLESS SCIENCE MISSION

JSS 1-3 | Soilless Learning Garden

Use scientific investigation to model a question about recirculating soilless operation. Quantify the response and defend a recommendation.

MISSION KIT system map, pH/EC/flow tools

EVIDENCE solution, flow or root-zone evidence

GM-378 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Model with system map, pH/EC/flow tools. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for solution, flow or root-zone evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Teacher supervises solution measurements. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Soilless Learning Garden, including the evidence that triggers the decision.

EVALUATE THE EVIDENCE

JSS 1-3 | Soilless Learning Garden

Use scientific investigation to evaluate a question about recirculating soilless operation. Quantify the response and defend a recommendation.

MISSION KIT system map, pH/EC/flow tools

EVIDENCE solution, flow or root-zone evidence

GM-379 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Evaluate with system map, pH/EC/flow tools. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for solution, flow or root-zone evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Teacher supervises solution measurements. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Soilless Learning Garden, including the evidence that triggers the decision.

RECOMMEND & TEST

JSS 1-3 | Soilless Learning Garden

Use scientific investigation to recommend a question about recirculating soilless operation. Quantify the response and defend a recommendation.

MISSION KIT system map, pH/EC/flow tools

EVIDENCE solution, flow or root-zone evidence

GM-380 - MISSION PATH

- DO** 1. Define the problem and hypothesis. 2. Specify variables, treatment/comparison and replicates. 3. Recommend with system map, pH/EC/flow tools. 4. Quantify the response. 5. Analyze variation or effect. 6. Conclude and recommend.
- PROVE** Submit the experimental layout, raw-data table, calculation/graph and conclusion for solution, flow or root-zone evidence.
- SUCCESS** Variables and units are defined, evidence is traceable, analysis is appropriate, and the recommendation states uncertainty.
- SAFETY** Teacher supervises solution measurements. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Translate the result into one management decision for the Soilless Learning Garden, including the evidence that triggers the decision.